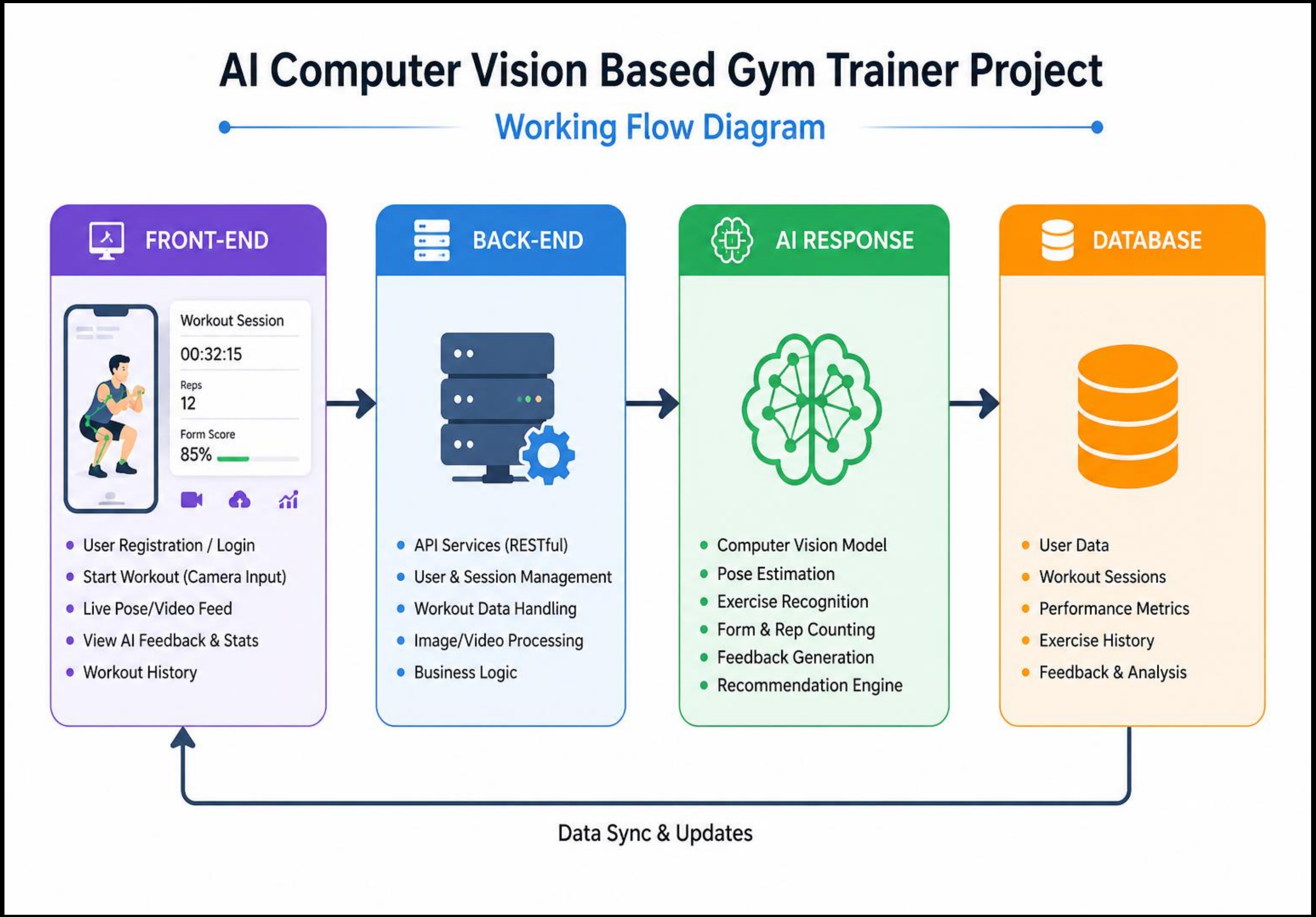
What is the Project?

Project Title:

Gym Tracker: An AI Computer Vision Based Gym Trainer.

- A smart fitness system using AI and computer vision
- Detects human body posture using camera
- Counts exercise repetitions automatically
- Provides real-time feedback (Like a virtual trainer)



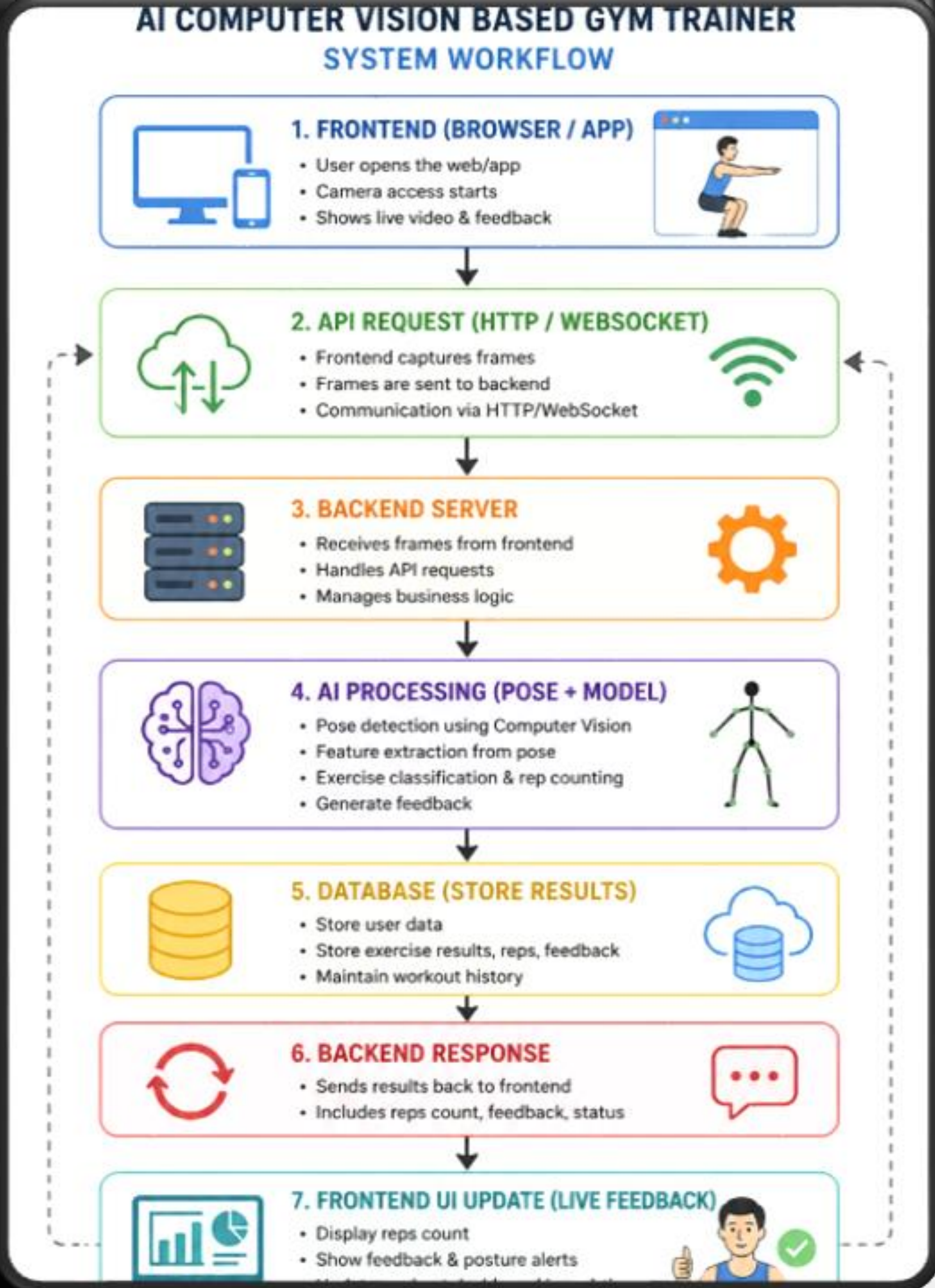
System Design (Architecture)

Bullet Points:

- Frontend: Camera + UI
- Backend: API + Processing
- AI Model: Prediction
- Database: Stores results

Design Of Complete Working Flow

- This system demonstrates the complete workflow of an AI-based gym trainer. The frontend captures live video and sends frames to the backend through API requests. The backend processes the data using AI models for pose detection and exercise classification. The results are stored in a database and sent back to the frontend, where real-time feedback, repetition counting, and posture correction are displayed to the user.



01 ECONOMIC IMPACT

- Reduces dependency on expensive personal trainers
- Affordable fitness solution accessible to a wider population
- Creates opportunities in AI-based fitness technology market
- Supports digital fitness startups and innovation

02 SOCIAL IMPACT

- Promotes home-based and flexible workouts
- Encourages a healthy lifestyle across different age groups
- Increases accessibility for people in remote or underserved areas
- Enhances user engagement through interactive fitness experience

03 POLITICAL IMPACT

- Supports government initiatives for public health awareness
- Aligns with digital transformation and smart technology adoption
- Can be integrated into national fitness and wellness programs
- Encourages development of AI-driven healthcare solutions

04 HEALTH IMPACT

- Promotes regular physical activity and fitness discipline
- Helps users maintain correct posture and reduce injury risk
- Provides real-time feedback similar to a personal trainer
- Contributes to prevention of lifestyle-related diseases

Ethical & Professional Responsibility

- Data Privacy & Security
- Ensure user video data is not stored without consent
- Protect personal information using secure systems
- Follow data protection standards and private policy.

- Fairness & Bias
- AI model should work for all body types, genders, and fitness levels
- Avoid biased training data
- Ensure equal performance across diverse users.

Accuracy & Reliability

•Provide correct posture detection and feedback
•Avoid misleading or incorrect exercise guidance
•Clearly mention system limitations to users.

Professional Responsibility

•Develop the system with transparency and honesty
•Ensure safe and responsible use of AI technology
•Continuously improve system performance and reliability.

TOOLS & TECHNOLOGIES USED

AI Computer Vision Based Gym Trainer Capstone Project

1. MEDIAPIPE

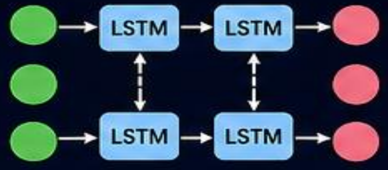
MediaPipe

- Real-time pose estimation and landmark detection
- Detects 33 key body landmarks
- Provides accurate 3D pose tracking
- Used for posture analysis and movement tracking

2. ANGLE LOGIC

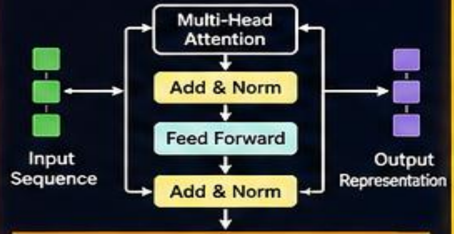
- Calculates joint angles using landmarks
- Evaluates posture correctness
- Counts repetitions based on angle thresholds
- Provides real-time feedback

3. BI-LSTM MODEL



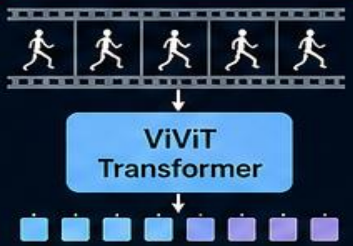
- Captures temporal dependencies in exercise sequences
- Understands forward and backward context
- Improves accuracy in repetition counting and exercise recognition
- Handles variable length sequences effectively

4. ENCODER TRANSFORMER MODEL



- Learns long-range dependencies
- Focuses on important features in the sequence
- Improves understanding of complex movements
- Generates rich feature representations

5. VIVIT TRANSFORMER MODEL



- Processes video as spatio-temporal tokens
- Captures both spatial and temporal features
- Highly effective for action recognition
- Enhances overall model performance

OTHER TECHNOLOGIES

Tool	Use
Python	Core Programming Language
OpenCV	Image Processing & Computer Vision
TensorFlow	Model Building & Training
Keras	Deep Learning API
NumPy	Numerical Computing
Pandas	Data Handling & Analysis
WebRTC	Real-time Video Streaming

TECHNOLOGY INTEGRATION

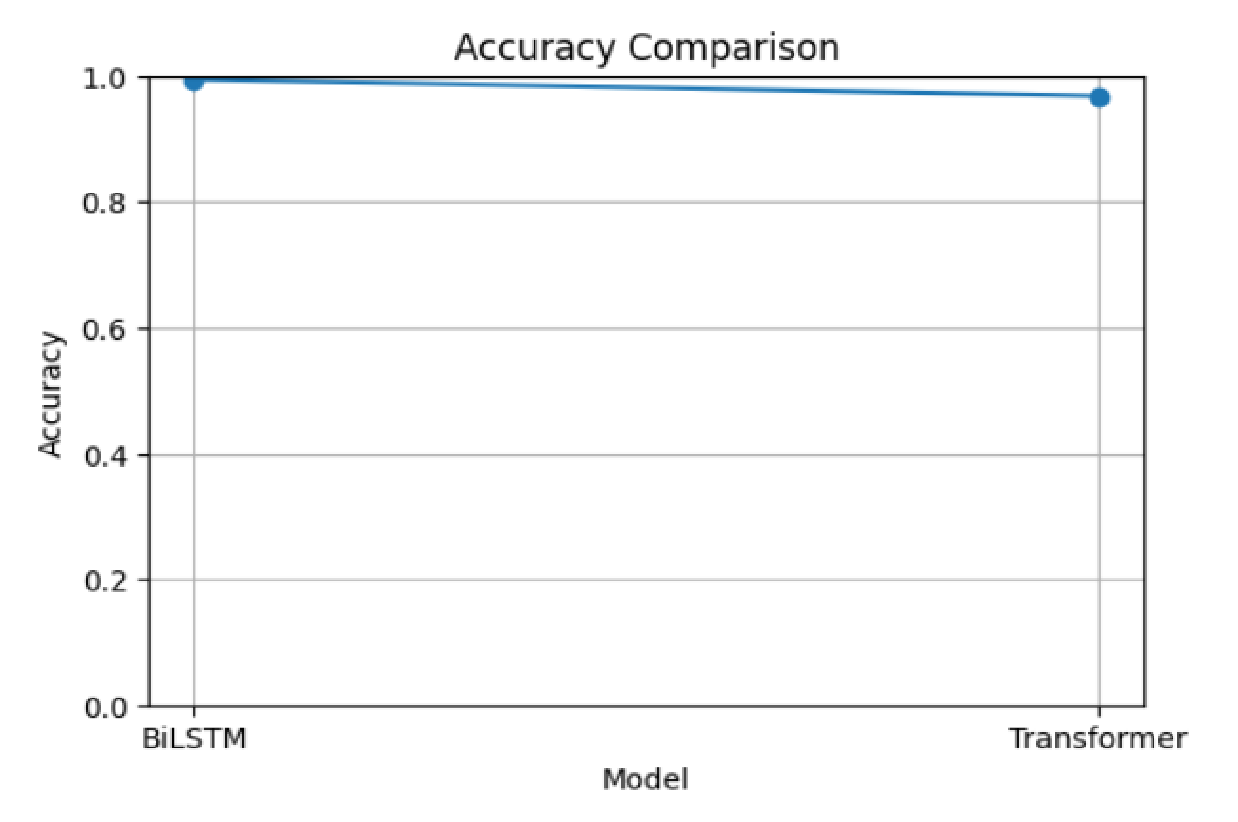
These tools and models work together to enable real-time pose estimation, exercise recognition, repetition counting, and intelligent feedback generation.

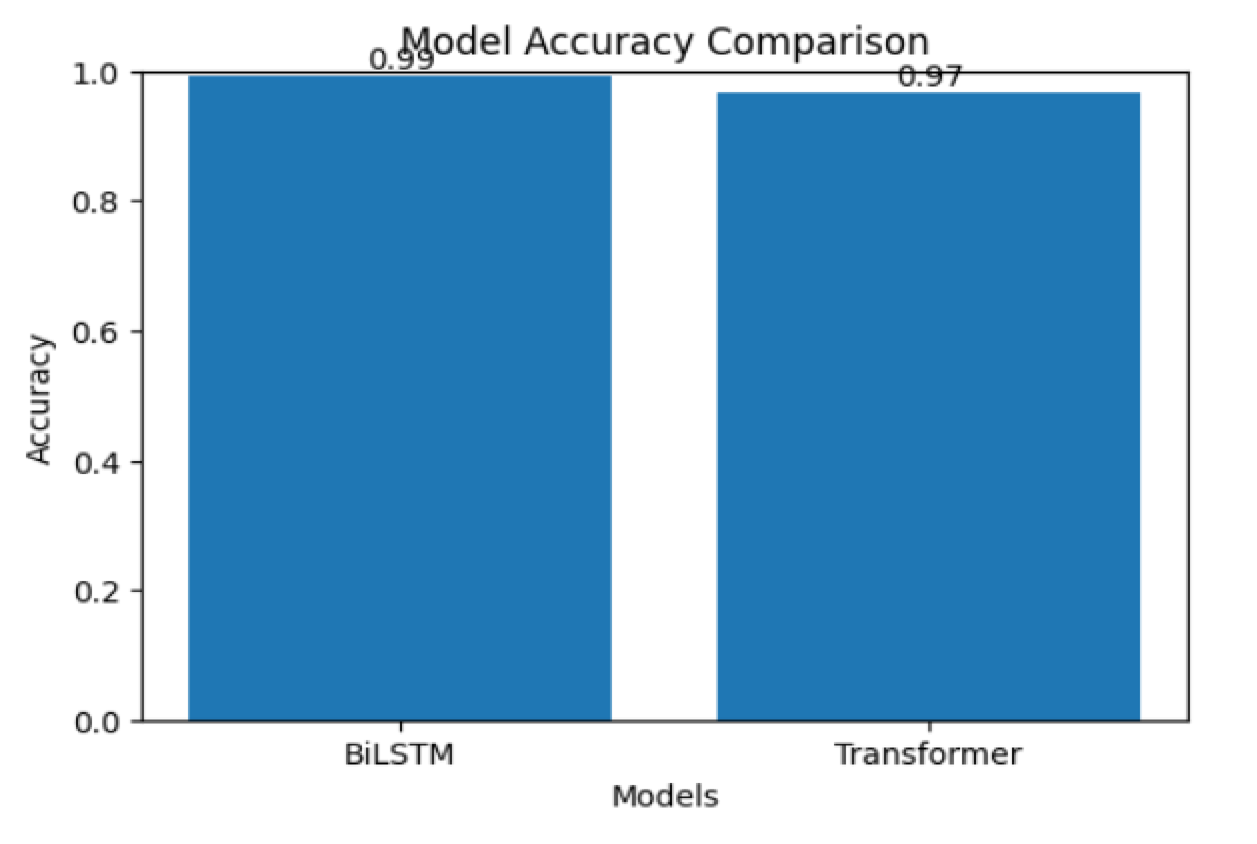
Results (Accuracy)

Model Used:

- Bi-LSTM Model
- Encoder Transformer Model
- ViViT Transformer Model

	Model	Input Shape	Parameters	Accuracy
0	BiLSTM	(30,78)	116644	0.994053
1	Transformer	(30,78)	99502	0.968062





Results (Code)

Core Training Technology Used:

01. MediaPipe Pose
02. ViViT (Transformer)
03. Encoder Transformer
04. Normal Transformer
05. Bi-LSTM
06. Angle Logic

MediaPipe Framework Called:

```
import cv2
import numpy as np
import mediapipe as mp

mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=1,
    smooth_landmarks=True,
    min_detection_confidence=0.6,
    min_tracking_confidence=0.6
)
```

ViViT (Transformer) Model:

```
from transformers import VivitForVideoClassification

model_name = "google/vivit-b-16x2-kinetics400"
model = VivitForVideoClassification.from_pretrained(model_name)

print("Model loaded successfully")
print("Number of labels:", model.config.num_labels)
```

```
config.json:  18.6k/? [00:00<00:00, 1.13MB/s]
pytorch_model.bin: 100%  356M/3
Loading weights: 100%  200/200
Model loaded successfully
Number of labels: 400
```